

Register No:

3 1 2 2 2 1 5 0 0 1 0 0 1

Sri Sivasubramaniya Nadar College of Engineering, Kalavakkam – 603 110.

(An Autonomous Institution, Affiliated to Anna University, Chennai)

B.E. / B.Tech. End Semester Theory Examinations, April / May 2024.

Sixth Semester

Computer Science and Engineering

UCS2603 THEORY OF COMPUTATION

(Regulations 2021)

Time: **Three Hours**Maximum: **100 Marks**

K1: Remembering

K2: Understanding

K3: Applying

K4: Analyzing

K5: Evaluating

CO1:	Construct mathematical proofs related to computation and finite automata for a given language (K3)
CO2:	Identify relationship between finite automata and regular expressions (K3)
CO3:	Distinguish different types of grammars, analyze relationship between language and context free grammar and construct PDA for CFG (K4)
CO4:	Construct Turing machine for a given language (K3)
CO5:	Explain the decidability or undecidability of various problems and make use of it (K3)

Part – A (5 × 2 = 10 Marks)

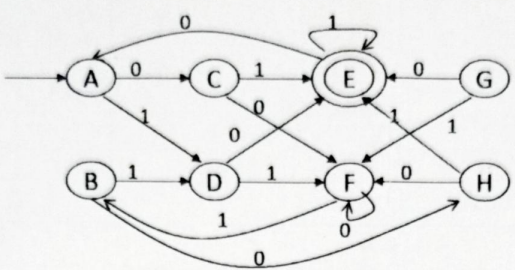
		KL	CO	PI
1.	Let A be a DFA and 'q' a particular state of A, such that $\delta(q, a) = q$ for all input symbols 'a'. Show by induction on the length of the input that for all input strings w, $\hat{\delta}(q, w) = q$.	K2	CO1	1.3.1
2.	Construct an NFA with ϵ - transitions for the given regular expression $00(0 + 1)^*$.	K3	CO2	1.4.1
3.	Construct context-free grammar for the following language: $\{a^i b^j c^k \mid i \neq j \text{ and } j \neq k\}$.	K3	CO3	1.4.1 2.1.3
4.	Compare Pushdown Automata and Turing Machine.	K2	CO4	1.4.1
5.	Prove that recursive languages are closed under complementation.	K2	CO5	1.4.1 2.1.3

Part – B (5 × 6 = 30 Marks)

		KL	CO	PI																
6.	Construct ϵ-closure of each state and its equivalent NFA for the following ϵ-NFA. <table><tr><td>δ</td><td>ϵ</td><td>0</td><td>1</td></tr><tr><td>$\rightarrow p$</td><td>$\{r\}$</td><td>$\{q\}$</td><td>$\{p, r\}$</td></tr><tr><td>q</td><td>$\emptyset$</td><td>$\{p\}$</td><td>$\emptyset$</td></tr><tr><td>$*r$</td><td>$\{p, q\}$</td><td>$\{r\}$</td><td>$\{p\}$</td></tr></table>	δ	ϵ	0	1	$\rightarrow p$	$\{r\}$	$\{q\}$	$\{p, r\}$	q	$\emptyset$	$\{p\}$	$\emptyset$	$*r$	$\{p, q\}$	$\{r\}$	$\{p\}$	K3	CO1	2.1.3, 13.3.1
δ	ϵ	0	1																	
$\rightarrow p$	$\{r\}$	$\{q\}$	$\{p, r\}$																	
q	$\emptyset$	$\{p\}$	$\emptyset$																	
$*r$	$\{p, q\}$	$\{r\}$	$\{p\}$																	
7.	State the pumping lemma for regular languages. Let $\Sigma = \{a, b\}$, prove the language $L = \{w \in \Sigma^* \mid n_a(w) < n_b(w)\}$ is not regular.	K3	CO2	2.1.3 2.4.1																
8.	Show that every regular language is a context-free language.	K2	CO3	1.4.1																
9.	List and explain the modifications of Turing Machine.	K2	CO4	2.4.1																
10.	Explain Post Correspondence Problem. Verify and explain whether there exists a solution for the Modified PCP given 2 lists $A = \{1, 111, 10, 0\}$ and $B = \{11, 1, 0111, 10\}$.	K2	CO5	2.1.3 13.3.1																

Part – C (5 × 12 = 60 Marks)

		KL	CO	PI
11.	(a) Construct a DFA that accepts the following language: $L = \{w \in (0, 1)^* \mid w \text{ ends in } 00\}$. 5 Marks (b) Design a DFA to accept the language $L = \{w \in (a, b, c)^* \mid w \text{ starts and ends with the same symbol}\}$. 7 Marks	K3	CO1	2.1.3 2.4.1 13.3.1
(Or)				
12.	Convert the following Non-deterministic Finite Automata M_1 to an equivalent Deterministic Finite Automata M_2 using the subset construction procedure. Prove that $L(M_1) = L(M_2)$. <pre> graph LR start(()) --> q0((q0)) q0 -- "a, b" --> q0 q0 -- "a" --> q1((q1)) q1 -- "b" --> q1 q1 -- "a" --> q2(((q2))) </pre>	K3	CO1	2.1.3 2.4.1 13.3.1

13.	Write the procedure to minimize a DFA. Construct the minimized DFA for the following DFA. 	K3	CO2	1.3.1 2.1.3 2.4.1 13.3.1
(Or)				
14.	Solve: Let $L_1 = \{w \in \{a, b\}^* \mid \text{begin with a and ends with b}\}$ and $L_2 = \{x \in \{a, b\}^* \mid \text{aaa is not a subword}\}$. If L_1 and L_2 are regular languages, then prove that $L_1 \cap L_2$ is regular with appropriate diagram.	K3	CO2	1.3.1 2.1.3 2.4.1 13.3.1
(Or)				
15.	(a) Define the Chomsky Hierarchy of Grammar (Mention rules form for each type of grammar and the restriction). 5 Marks (b) Convert the following grammar into a reduced grammar. 7 Marks $\begin{aligned} S &\rightarrow aA \mid aBB, \\ A &\rightarrow aaA \mid \epsilon, \\ B &\rightarrow bB \mid bbC, \\ C &\rightarrow B \end{aligned}$	K3	CO3	1.4.1 2.1.3 2.4.1 13.3.1
(Or)				
16.	Convert the PDA $M = (\{q\}, \{a, b\}, \{Z, A, B\}, \delta, q, Z, \varphi)$ to an equivalent CFG, if δ is given by: $\begin{aligned} \delta(q, a, Z) &= (q, AZ), \\ \delta(q, b, Z) &= (q, BZ), \\ \delta(q, a, A) &= (q, AA), \\ \delta(q, b, B) &= (q, BB), \\ \delta(q, a, B) &= (q, \epsilon), \\ \delta(q, b, A) &= (q, \epsilon), \end{aligned}$	K3	CO3	1.4.1 2.1.3 2.4.1 13.3.1

	$\delta(q, \epsilon, Z) = (q, \epsilon)$			
17.	Construct a Turing Machine to perform subtraction of 2 numbers. $f(m, n) = m - n \text{ if } m > n$ $f(m, n) = B \text{ if } m \leq n$ Show the validation of $w = 0010$.	K3	CO4	1.4.1 2.4.1 13.3.1 13.4.1
(Or)				
18.	Consider the two track Turing Machine M as a 7-tuple $(Q, \Sigma, \Gamma, \delta, q_0, B, F)$. Using M , design a two track Turing Machine for the Language $L = \{ 0^k 1^k \mid k \geq 1 \}$.	K3	CO4	1.4.1 2.4.1 13.3.1 13.4.1
19.	Prove the following: a) If A and B are decidable, then $A \cup B$ is decidable. b) If A and B are undecidable, then $A \cup B$ is undecidable. c) If A and B are non recursively enumerable, then $A \cup B$ is non recursively enumerable.	K3	CO5	2.1.3 2.4.1 13.3.1
(Or)				
20.	(a) Consider the Turing machine $M = (\{q_0, q_1, q_2, q_3\}, \{0, 1\}, \{G, 1, B\}, \delta, q_0, B, \{q_3\})$ with δ consisting of the following sets of rules: $\delta(q_0, 0) = \{(q_0, 0, R)\},$ $\delta(q_0, 1) = \{(q_1, 1, R)\},$ $\delta(q_1, 1) = \{(q_2, 1, R)\},$ $\delta(q_1, 0) = \{(q_1, 0, R)\},$ $\delta(q_2, 0) = \{(q_2, 0, R)\},$ $\delta(q_2, 1) = \{(q_3, 1, R)\},$ $\delta(q_3, 0) = \{(q_3, 0, R)\},$ $\delta(q_3, 1) = \{(q_0, 1, R)\}.$ Informally but clearly describe the language $L(M)$. Construct a binary encoding for M .	K3	CO5	2.1.3 2.4.1 13.3.1